AI Model Gateway Customer Guide

A simple reference for customers who want a multi-model AI platform direction, but may not have a large technical team yet.

Audience	Business and product teams with basic technical awareness
Prototype	Local AI Model Gateway with Qwen / DashScope first
Main idea	One customer API, many model providers behind the scenes
Demo mode	Mock mode first, live Qwen mode only when needed

Executive Summary

An AI Model Gateway lets customers call one simple API while the platform manages model routing, provider differences, usage control, and future fallback logic. The first prototype uses Alibaba Cloud Model Studio / Qwen because it is available now and keeps testing cost low.

- Start with Qwen and mock mode before adding more paid providers.
- Use an OpenAI-compatible API so customers can understand the integration quickly.
- Show routing visually: smart-fast is a customer-facing name that maps to qwen-plus.
- Keep enterprise API Gateway features optional until customer demand is clear.

Layered Architecture

Layer	What It Means	Example
1. Customer Systems	The customer's app, backend, or agent	Website, chatbot, internal tool
2. One Simple API	One standard API format for customers	/v1/chat/completions
3. Gateway Entry Control	Controls who can call and how much they can use	API keys, limits, logs
4. Model Control	Chooses the public model and route	smart-fast -> qwen-plus
5. Provider Adapters	Translates request and response formats	DashScope adapter
6. Model Providers	The real upstream AI model provider	Alibaba Cloud Qwen

How Route Simulation Works

The dashboard explains routing without spending money. In mock mode, the gateway reads the requested model, checks the registry, maps the public model to a real upstream model, and returns a simulated response with a route trace.

```
Customer request: model = smart-fast
Registry lookup: smart-fast maps to qwen-plus
Provider adapter: prepare DashScope-compatible request
Mock response: show the route trace without calling Alibaba Cloud
```

What The Current Prototype Includes

- Visual dashboard at the gateway root URL.
- OpenAI-compatible /v1/chat/completions endpoint.
- Customer API key authentication with Bearer token.
- Basic in-memory request limit.
- Model registry in model_registry.json.
- SQLite request and usage records for restart-safe demo data.
- Bring Your Own Key mapping through environment variables.
- Provider adapter scaffolds for OpenAI-compatible APIs and Claude-style APIs.
- Mock mode for demos and planning.
- Live Qwen mode when DASHSCOPE_API_KEY is available.

Main Technical Difficulties

Difficulty	Why It Matters	Prototype Status
Provider format differences	Each provider may use different request and response details.	Handled first for DashScope-compatible chat
Streaming	Chat UIs and agents often need incremental tokens.	Mock streaming included; live provider differences
Tool calling	OpenAI, Claude, and Qwen may differ in tool format.	Basic normalization scaffold
Token usage	Billing and quota require reliable usage data.	Stored in SQLite
Fallback	Retrying another model needs clear business rules.	Mock fallback test included
Customer key management	Each customer needs limits, logs, and access control.	Minimum version plus BYOK mapping
Cost control	Different models have different prices and limits.	Future usage and billing layer

Recommended Roadmap

Phase	Goal	Result
1. Current prototype	Qwen-only gateway with mock dashboard	Customer can understand the concept
2. Live Qwen demo	Use Model Studio API key for real chat	Validate latency and response quality
3. Gateway controls	Persist logs, usage, model access rules, and BYOK mapping	Better customer management
4. More providers	Enable OpenAI, Claude, Xiaomi adapters	More complete OpenRouter-like direction
5. Production	Database, secrets, streaming, fallback, billing	Enterprise-ready platform path

Reference Documents

OpenRouter API Reference: https://openrouter.ai/docs/api/reference/overview/

OpenRouter Authentication: https://openrouter.ai/docs/api-keys

OpenRouter Limits: https://openrouter.ai/docs/api-reference/limits/

OpenRouter Models API: https://openrouter.ai/docs/api/api-reference/models/get-models

OpenRouter Model Fallbacks: https://openrouter.ai/docs/guides/routing/model-fallbacks

OpenRouter Provider Routing: https://openrouter.ai/docs/guides/routing/provider-selection/

OpenRouter BYOK: https://openrouter.ai/docs/use-cases/byok/

Alibaba Cloud Model Studio DashScope API Reference: https://www.alibabacloud.com/help/doc-detail/3016809.html

Final Recommendation

Start small and explain clearly. Use the visual dashboard to show the customer-facing API, the routing layer, and the provider adapter. Use mock mode for discussion and live Qwen mode only when real model testing is needed. Add more providers only after the customer understands the value of the gateway.